D. Interventions for common physiological symptoms

Background

Women's bodies undergo substantial changes during pregnancy, which are brought about by both hormonal and mechanical effects. These changes lead to a variety of common symptoms – including nausea and vomiting, low back and pelvic pain, heartburn, varicose veins, constipation and leg cramps – that in some women cause severe discomfort and negatively affects their pregnancy experience. In general, symptoms associated with mechanical effects, e.g. pelvic pain, heartburn and varicose veins, often worsen as pregnancy progresses.

Symptoms of nausea and vomiting are experienced by approximately 70% of pregnant women and usually occur in the first trimester of pregnancy *(156)*; however, approximately 20% of women may experience nausea and vomiting beyond 20 weeks of gestation *(157)*. Low back and pelvic pain is estimated to occur in half of pregnant women, 8% of whom experience severe disability *(158)*. Symptoms of heartburn occur in two thirds of pregnant women, and may be worse after eating and lying down *(159)*. Varicose veins usually occur in the legs, but can also occur in the vulva and rectum, and may be associated with pain, night cramps, aching and heaviness, and worsen with long periods of standing *(160)*. Constipation can be very troublesome and may be complicated by haemorrhoids *(161)*. Leg cramps often occur at night and can be very painful, affecting sleep and daily activities *(162)*. Suggested approaches to manage common physiological symptoms include a variety of non-pharmacological and pharmacological options and the GDG considered the evidence and other relevant information on these approaches.

Women's values:

A scoping review of what women want from ANC and what outcomes they value informed the ANC guideline *(13)*. Evidence showed that women from high-, medium- and low-resource settings valued having a positive pregnancy experience. This included woman-centred advice and treatment for common physiological symptoms (high confidence in the evidence). In many LMICs, this also included support and respect for women's use of alternative or traditional approaches to the diagnosis and treatment of common pregnancy-related symptoms (moderate confidence in the evidence).

D.1: Interventions for nausea and vomiting

RECOMMENDATION D.1: Ginger, chamomile, vitamin B6 and/or acupuncture are recommended for the relief of nausea in early pregnancy, based on a woman's preferences and available options. *(Recommended)*

Remarks

- In the absence of stronger evidence, the GDG agreed that these non-pharmacological options are unlikely to have harmful effects on mother and baby.
- Women should be informed that symptoms of nausea and vomiting usually resolve in the second half of pregnancy.
- Pharmacological treatments for nausea and vomiting, such as doxylamine and metoclopramide, should be reserved for those pregnant women experiencing distressing symptoms that are not relieved by non-pharmacological options, under the supervision of a medical doctor.

Summary of evidence and considerations

Effects of interventions for nausea and vomiting compared with other, no or placebo interventions (EB Table D.1)

The evidence on the effects of various interventions for nausea and vomiting in pregnancy was derived from a Cochrane systematic review *(157)*. The review included 41 trials involving 5449 women in whom a wide variety of pharmacological and non-pharmacological interventions were evaluated. Trials were conducted in a variety of HICs and LMICs, and most included pregnant women at less than 16 weeks of gestation with mild to moderate nausea and vomiting. Alternative therapies and non-pharmacological agents evaluated included acupuncture, acupressure, vitamin B6, ginger, chamomile, mint oil and lemon oil. Pharmacological agents included antihistamines, phenothiazines, dopamine-receptor antagonists and serotonin 5-HT3 receptor antagonists. Due to heterogeneity among the types of interventions and reporting of outcomes, reviewers were seldom able to pool data. The primary outcome of all interventions was maternal relief from symptoms (usually measured using the Rhodes Index), and perinatal outcomes relevant to this guideline were rarely reported.

Non-pharmacological agents versus placebo or no treatment

Ten trials evaluated non-pharmacological interventions including ginger (prepared as syrup, capsules or powder within biscuits) (7 trials from the Islamic Republic of Iran, Pakistan, Thailand and the USA involving 578 participants), lemon oil (one Iranian study, 100 participants), mint oil (one Iranian study, 60 participants), chamomile (one Iranian study, 105 participants), and vitamin B6 interventions (two studies in Thailand and the USA; 416 participants) compared with no treatment or placebo.

Ginger: Low-certainty evidence from several small individual studies suggests that ginger may relieve symptoms of nausea and vomiting. A study from Pakistan found that ginger reduced nausea symptom scores (68 women; MD: 1.38 lower on day 3, 95% CI: 0.03–2.73 lower), and vomiting symptom scores (64 women; MD: 1.14 lower, 95% CI: 0.37–1.91 lower), and an Iranian study showed improvements in nausea and vomiting symptom scores on day 7 in women taking ginger supplements compared with placebo (95 women; MD: 4.19 lower, 95% CI: 1.73–6.65 lower). Data from the studies in Thailand and the USA showed a similar direction of effect on nausea symptoms in favour of ginger.

Lemon oil: Low-certainty evidence from one small Iranian study suggests that lemon oil may make little or no difference to nausea and vomiting symptom scores (100 women; MD: 0.46 lower on day 3, 95% CI: 1.27 lower to 0.35 higher), or to maternal satisfaction (the number of women satisfied with treatment) (1 trial, 100 women; RR: 1.47, 95% CI: 0.91–2.37).

Mint oil: The evidence on mint oil's ability to relieve symptoms of nausea and vomiting is of very low certainty.

Chamomile: Low-certainty evidence from one small study suggests that chamomile may reduce nausea and vomiting symptoms scores (70 women; MD: 5.74 lower, 95% CI: 3.17–8.31 lower).

Vitamin B6 (pyridoxine): Moderate-certainty evidence from two trials (one used 25 mg oral vitamin B6 8-hourly for 3 days, the other used 10 mg oral vitamin B6 8-hourly for 5 days) shows that vitamin B6 probably reduces nausea symptoms scores (388 women, trials measured the change in nausea scores from baseline to day 3; MD: 0.92 higher score change, 95% 0.4–1.44 higher), but low-certainty evidence suggests that it may have little or no effect on vomiting (2 trials, 392 women; RR: 0.76, 95% CI: 0.35–1.66).

Acupuncture and acupressure versus placebo or no treatment

Five studies (601 participants) evaluated P6 (inner forearm) acupressure versus placebo, one Thai study (91 participants) evaluated auricular acupressure (round magnetic balls used as ear pellets) versus no treatment, one study in the USA (230 participants) evaluated P6 acustimulation therapy (nerve stimulation at the P6 acupuncture point) versus placebo, and a four-arm Australian study (593 women) evaluated traditional Chinese acupuncture or P6 acupuncture versus P6 placebo acupuncture or no intervention.

Low-certainty evidence suggests that P6 acupressure may reduce nausea symptom scores (100 women; MD: 1.7 lower, 95% CI: 0.99–2.41 lower) and reduce the number of vomiting episodes (MD: 0.9 lower, 95% CI: 0.74–1.06 lower). Low-certainty evidence suggests that auricular acupressure may also reduce nausea symptom scores (91 women; MD: 3.6 lower, 95% CI: 0.58–6.62 lower), as may traditional Chinese acupuncture (296 women; MD: 0.7 lower, 95% CI: 0.04–1.36 lower). Low-certainty evidence suggests that P6 acupuncture may make little or no difference to mean nausea scores compared with P6 placebo acupuncture (296 women; MD: 0.3 lower, 95% CI: 1.0 lower to 0.4 higher).

Pharmacological agents versus placebo

One study evaluated an antihistamine (doxylamine) and another evaluated a dopamine-receptor antagonist (metoclopramide). Certain other drugs evaluated in the review (hydroxyzine, thiethylperazine and fluphenazine) are from old studies and these drugs are no longer used in pregnant women due to safety concerns.

Moderate-certainty evidence suggests that doxylamine plus vitamin B6 probably reduces nausea and vomiting symptom scores compared with placebo (1 study, 256 women; MD: 0.9 lower on day 15, 95% CI: 0.25–1.55 lower). Low-certainty evidence from this study suggests that there may be little or no difference in headache (256 women; RR: 0.81, 95% CI: 0.45–1.48) or drowsiness (256 women; RR: 1.21, 95% CI: 0.64–2.27) between doxylamine plus vitamin B6 and placebo.

Low-certainty evidence on metoclopramide (10 mg) suggests that this agent may reduce nausea symptom scores (1 trial, 68 women; MD: 2.94 lower on day 3, 95% CI: 1.33–4.55 lower). There was no side-effect data on metoclopramide in the review.

No studies compared ondansetron (a 5HT3 receptor antagonist) with placebo. Two small studies compared ondansetron with metoclopramide and doxylamine, respectively, but evidence on relative effects was uncertain.

Additional considerations

- Low-certainty evidence from single studies comparing different non-pharmacological interventions with each other – namely acupuncture plus vitamin B6 versus P6 acupuncture plus placebo (66 participants), traditional acupuncture and P6 acupuncture (296 participants), ginger versus chamomile (70 participants), P6 acupuncture versus ginger (98 participants), and ginger versus vitamin B6 (123 participants) – suggests there may be little or no difference in effects on relief of nausea symptoms.
- Low-certainty evidence suggests that there may be little or no difference between ginger and metoclopramide on nausea symptom scores (1 trial, 68 women; MD: 1.56 higher, 95% 0.22 lower to 3.34 higher) or vomiting symptom scores (68 women; MD: 0.33 higher, 95% CI: 0.69 lower to 1.35 higher) on day 3 after the intervention.
- Side-effects and safety of pharmacological agents were poorly reported in the included studies. However, drowsiness is a common side-effect of various antihistamines used to treat nausea and vomiting.
- Metoclopramide is generally not recommended in the first trimester of pregnancy, but is widely used *(163)*. A study of over 81 700 singleton births in Israel reported that they found no statistically significant differences in the risk of major congenital malformations, low birth weight, preterm birth or perinatal death between neonates exposed (3458 neonates) and not exposed to metoclopramide in the first trimester of gestation.

Values

See "Women's values" at the beginning of section 3.D: Background (p. 74).

Resources

Costs associated with non-pharmacological remedies vary. Acupuncture requires professional training and skills and is probably associated with higher costs. Vitamin B6 (pyridoxine hydrochloride tablets) could cost about US$ 2.50 for 90 × 10 mg tablets *(74)*.

Equity

The impact on equity is not known.

Acceptability

Qualitative evidence from a range of LMICs suggests that women may be more likely to turn to traditional healers, herbal remedies or traditional birth attendants (TBAs) to treat these symptoms (moderate confidence in the evidence) *(22)*. In addition, evidence from a diverse range of settings indicates that while women generally appreciate the interventions and information provided during antenatal visits, they are less likely to engage with services if their beliefs, traditions and socioeconomic circumstances are ignored or overlooked by health-care providers and/or policy-makers (high confidence in the evidence). This may be particularly pertinent for acupuncture or acupressure, which may be culturally alien and/or poorly understood in certain contexts.

Feasibility

A lack of suitably trained staff may limit feasibility of certain interventions (high confidence in the evidence) *(45)*.

D.2: Interventions for heartburn

RECOMMENDATION D.2: Advice on diet and lifestyle is recommended to prevent and relieve heartburn in pregnancy. Antacid preparations can be offered to women with troublesome symptoms that are not relieved by lifestyle modification. *(Recommended)*

Remarks

- Lifestyle advice to prevent and relieve symptoms of heartburn includes avoidance of large, fatty meals and alcohol, cessation of smoking, and raising the head of the bed to sleep.
- The GDG agreed that antacids, such as magnesium carbonate and aluminium hydroxide preparations, are probably unlikely to cause harm in recommended dosages.
- There is no evidence that preparations containing more than one antacid are better than simpler preparations.
- Antacids may impair absorption of other drugs *(164)*, and therefore should not be taken within two hours of iron and folic acid supplements.

Summary of evidence and considerations

Effects of interventions for heartburn compared with other, no or placebo interventions (EB Table D.2)

The evidence on the effects of various interventions for heartburn in pregnancy comes from a Cochrane review that included nine trials involving 725 pregnant women with heartburn; however, only four trials (358 women) contributed data *(159)*. One of these, from the 1960s, evaluated intramuscular prostigmine, which is no longer used, therefore these data were not considered for the guideline. The three remaining studies conducted in Brazil, Italy and the USA evaluated a magnesium hydroxide–aluminium hydroxide–simeticone complex versus placebo (156 women), sucrulfate (aluminium hydroxide and sulfated sucrose) versus advice on diet and lifestyle changes (66 women), and acupuncture versus no treatment (36 women). Evidence on symptom relief was generally assessed to be of low to very low certainty and no perinatal outcomes relevant to this guideline were reported. Evidence on side-effects for all comparisons was assessed as being of very low certainty.

Pharmacological interventions versus placebo

Low-certainty evidence suggests that complete relief from heartburn may occur more frequently with magnesium hydroxide–aluminium hydroxide–simethicone liquid and tablets than placebo (156 women; RR: 2.04, 95% CI: 1.44–2.89).

Pharmacological interventions versus advice on diet and lifestyle changes

Low-certainty evidence suggests that complete relief from heartburn may occur more frequently with sucralfate than with advice on diet and lifestyle changes (65 women; RR: 2.41, 95% CI: 1.42–4.07).

Acupuncture versus no treatment

Data on relief of heartburn was not available in the review for this comparison. Low-certainty evidence suggests that weekly acupuncture in pregnant women with heartburn may improve the ability to sleep (36 women; RR: 2.80, 95% CI: 1.14–6.86) and eat (36 women; RR: 2.40, 95% CI: 1.11–5.18), a proxy outcome for maternal satisfaction.

Additional considerations

- Heartburn during pregnancy is a common problem that can be self-treated with over-the-counter products containing antacids such as magnesium carbonate, aluminium hydroxide or calcium carbonate.
- The Cochrane review found no evidence on prescription drugs for heartburn, such as omeprazole and ranitidine, which are not known to be harmful in pregnancy *(159)*.